

# Bottlenecks: Sectoral Imbalances and the US Productivity Slowdown

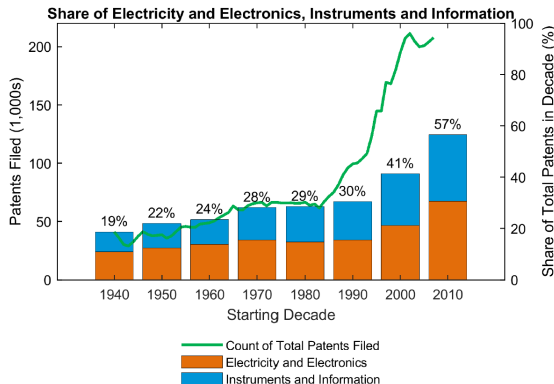
Acemoglu, Autor & Patterson (2020)

Presentation by Diva Astinova

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# Motivation: Macroeconomic Puzzle

- Large advances in ICT and electronics since the 1980s...

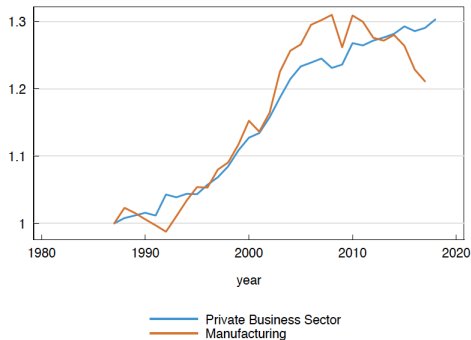


- Verge of a new age of abundance, or “singularity” (Kurzweil (2005), Diamandis and Kotler (2012), and Bostrom)

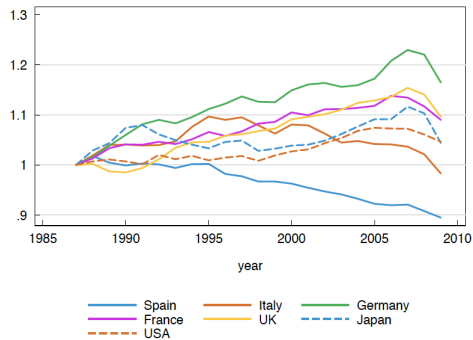
# Motivation: Macroeconomic Puzzle

- ... But TFP growth slowdown across industrialized nations in the last decades

(a) United States



(b) Select OECD Countries



- Age of slower growth (Cowen 2011, Gordon 2017)

# Preview of Results

This paper attempts to reconcile these two seemingly contradictory facts

- **Main Hypothesis:** Uneven technological advances create *endogenous bottlenecks* that hold back aggregate productivity
- **Model:** New technologies require simultaneous improvements in several inputs when advancements are *complements*
- **Empirical Evidence:** Greater dispersion in an industry's suppliers' TFP growth has large negative effect on its own TFP growth

## Illustrative Example: GPS

- Before GPS, navigation used sight-lines and optical instruments
- 1978: GPS satellites launched by military, providing geolocation and date-and-time information (atomic-level time-keeping)
- 1983: GPS opened for civilian use
- Subsequent innovations relying on GPS:
  - Precision agriculture
  - Weather prediction
  - Consumer-facing technologies (ride hailing, targeted advertising, etc.)

# Model - Production

- Model links sector  $i$ 's productivity to the productivity distribution of its suppliers
- CRS Cobb-Douglas production sector  $i = (1, \dots, N)$  at time  $t$

$$Y_{it} = B_i A_{it} L_{it}^{\left(1 - \sum_{j \in S_i} \alpha_{ij}\right)} \prod_{j \in S_i} X_{ijt}^{\alpha_{ij}} \quad (1)$$

- $L_{it}$  = labour
- $A_{it}$  = sector-specific productivity
- $X_{ijt}$  = vector of inputs necessary for production from sector  $j$
- $S_i$  = time invariant set of input sectors
- $\alpha_{ij}$  = input share of industry  $j$  in sector  $i$
- $B_i$  = normalizing constant

# Model - Technological Advances as Quality Improvements

- *Quality Ladder* Structure

$$A_{it} = \lambda^{n_{it}}$$

- Depends on # innovations ( $n_{it}$ ) that sector  $i$  has had in the past
- Each innovation increases productivity by a factor of  $\lambda > 1$

## Model - Arrival Rate of Innovation

- *Quality Ladder* Structure

$$A_{it} = \lambda^{n_{it}} \quad (2)$$

- **Critical Assumption:** Arrival rate of innovation ( $\phi_{it}$ ) depends on the technological advancement of inputs

$$\phi_{it} = H\left(\sum_{j \in S_i} \alpha_{ij} h(A_{jt})\right) \quad (3)$$

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- *Concave  $h$* : innovation across input sectors = **complements**
  - Technology of ALL inputs matters for innovation
  - More dispersion for input technology hinders innovation
- *Convex  $h$* : innovation across input sectors = *substitutes*
  - Most advanced inputs determine innovation

# Deriving Estimating Equations from Model

- Take 2<sup>nd</sup> order Taylor expansion of  $\phi_{it}$  around mean & 1<sup>st</sup> order expansion of  $H(\cdot)$  around 0:

$$\phi_{it} = H\left(\sum_{j \in S_i} \alpha_{ij} h(A_{jt})\right) \approx H\left[\alpha_{ij} h(\bar{A}_{it}) + h''(\bar{A}_{it}) \times \text{Var}(\{\alpha_{ij} A_{jt}\}_{j \in S_i})\right]$$

$$\phi_{it} \approx \eta_{mean}^i(\bar{A}_{it}) + \eta_{variance}^i \times \text{Var}(\{\alpha_{ij} A_{jt}\}_{j \in S_i})$$

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- $\bar{A}_{it} = \sum_{j \in S_i} \alpha_{ij} A_{jt}$ : (cost-share weighted) mean productivities of inputs to sector  $i$
- $\text{Var}(\{\alpha_{ij} A_{jt}\}_{j \in S_i})$ : (weighted) variance of productivity of inputs to sector  $i$

# Prediction of Estimating Equations from Model

$$\phi_{it} \approx \eta_{mean}^i(\bar{A}_{it}) + \eta_{variance}^i \times \text{Var}(\{\alpha_{ij}A_{jt}\}_{j \in S_i}) \quad (4)$$

- $\eta_{mean}^i$ : mean technological advances across inputs
- $\eta_{variance}^i$ : dispersion of innovations across inputs (mean-centered)
- **For increasing and concave h**
  - $\eta_{mean}^i = H'(0)h'(\bar{A}_{it}) > 0$        $\eta_{variance}^i = H'(0)h''(\bar{A}_{it}) < 0$

# Main Empirical Specification

$$\Delta TFP_{it} = \beta_{mean} \sum_{j \in S_i} \alpha_{ij,t-1} \Delta TFP_{jt} + \beta_{variance} Var(\Delta TFP_{jt}) + \delta_i + \delta_t + \varepsilon_{it} \quad (5)$$

$$\text{where } Var(\Delta TFP_{jt}) = \sum_{j \in S_i} \alpha_{ij,t-1} \left( \Delta TFP_{jt} - \sum_{j \in S_i} \alpha_{ij,t-1} \Delta TFP_{jt} \right)^2$$

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1. TFP growth is measure of innovation
2. 5-year TFP growth, not TFP levels

- Hypothesis for Sectoral Bottlenecks

$$H_0 : \beta_{mean} > 0 \qquad \beta_{variance} < 0$$

# Data

- **Input-Output Linkages:** detailed five year input-output tables from 1997 - 2002
- **Manufacturing Productivity (US):** 462 NAICS industries, NBER-CES Manufacturing Database
- **Non-Manufacturing Productivity (US):** 42 NAICS industries, BLS Multifactor Productivity Stats
- **International Productivity (OECD):** 30 NAICS industries, KLEMS
- **Patents:** industry-level from 1976-2006
- **Patent Citation Network:** Acemoglu et. al., 2016

# Main Results

- Variance in TFP growth of inputs drags down a sector's own TFP growth
- Dispersion of TFP growth across industries is significant!

# Bottleneck patterns present across all sectors

|                     | (1)              | (2)               | (3)               | (4)              | (5)               | (6)               |
|---------------------|------------------|-------------------|-------------------|------------------|-------------------|-------------------|
|                     | Manufacturing    |                   |                   | All Industries   |                   |                   |
| Input Average       | 0.425<br>(0.139) | 0.810<br>(0.130)  | 0.676<br>(0.170)  | 0.343<br>(0.178) | 0.915<br>(0.161)  | 0.636<br>(0.183)  |
| Input Variance      |                  | -0.744<br>(0.121) |                   |                  | -0.905<br>(0.158) |                   |
| Input Bottom Decile |                  |                   | 0.059<br>(0.113)  |                  |                   | 0.164<br>(0.099)  |
| Input Top Decile    |                  |                   | -0.110<br>(0.033) |                  |                   | -0.117<br>(0.034) |
| Observations        | 2772             | 2772              | 2772              | 2016             | 2016              | 2016              |
| R-Squared           | 0.108            | 0.133             | 0.118             | 0.079            | 0.102             | 0.090             |

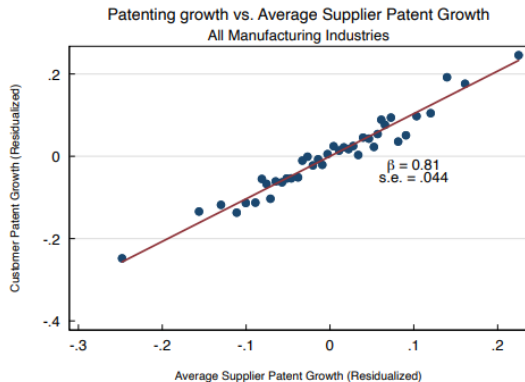
Notes: All regressions include year fixed effects. Columns 1-3 include stacked 5-year changes from 1977-2007 and columns 4-6 include include stacked 5-year changes from 1987-2007. Standard errors are clustered at the industry level. Industries are unweighted.

## Do these patterns capture innovation?

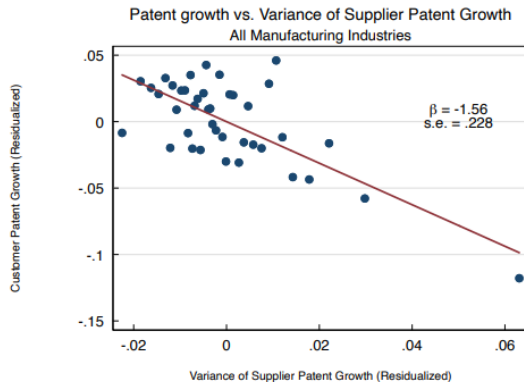
1. Driven by TFP, **not** by changes to prices or quantities
2. Bottleneck patterns appear using patents as measure of innovation instead of  $\Delta$ TFP
3. Bottleneck patterns appear using patent citation networks instead of TFP growth
4. Same patterns in other industrialized OECD countries outside the US

# Bottleneck patterns with patenting

- Analogous bottleneck result when measuring innovation growth with patents



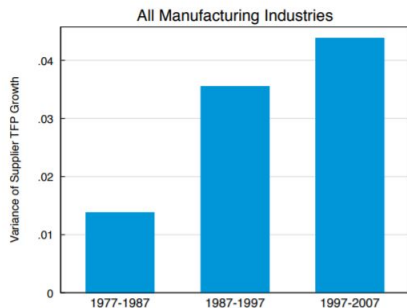
$$\beta_{mean} > 0$$



$$\beta_{variance} < 0$$

## Result: Dispersion of supplier TFP has grown over time

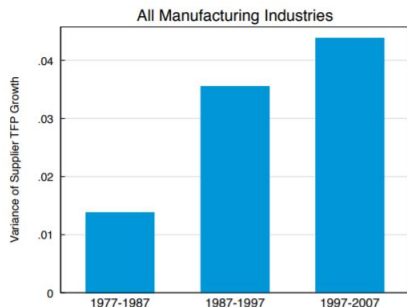
- Especially true in manufacturing



# Result: Dispersion of supplier TFP has grown over time

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At same time, productivity slowdown



↑ TFP variance across input suppliers accounts for:

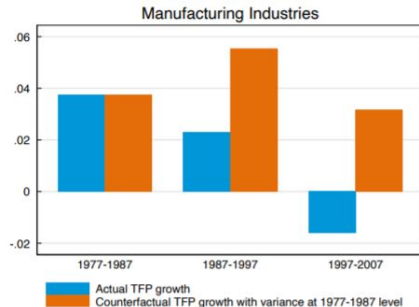
- Substantial part of agg productivity slowdown
- Nearly all ↓ fall of productivity in manufacturing

# Counterfactual 1 - Manufacturing Productivity

Keep  $Var(\Delta TFP)$  constant at 1977-1987 level

## Results

- Manufacturing productivity would grow **2x faster** in 1997-2007 than in data
- Manufacturing TFP growth from 1997-2007 would be + 3.1% under counterfactual instead of -1.6% in data

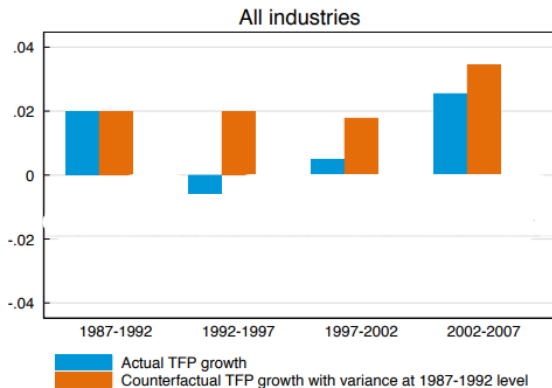


## Counterfactual 2 - Productivity Across All Sectors

Keep  $Var(\Delta TFP)$  constant at 1987-1992 level

### Results

- Aggregate productivity would have grown **2 pp faster** on average from 1992-2007 than in data



## Counterfactual 3 - Redistributing Productivity Growth

- Hold average TFP growth constant
- Redistribute 20 pp of TFP growth from 10 fastest-growing sectors to bottom 50% of industries

### Result

- ↑ Increases aggregate TFP by over 1 pp

Larger total productivity when technological progress is **balanced** across sectors

# Examples of Bottleneck & Leading Industries

*Leading Industries:* biggest positive effect on variance of inputs when  $\uparrow TFP_i$  by 10%

- Semiconductors
- Electronic Computers
- Iron and Steel Mills
- Computer Storage Devices

*Bottleneck Industries:* biggest negative effect on variance of inputs when  $\uparrow TFP_i$  by 10%

- Petroleum Refineries
- Pharmaceutical Preparation
- Turbine and Turbine Generator Units
- Printed Circuit Assembly

# Robustness

Negative coefficient of variance supplier TFP growth are robust to:

- i) **Auto-Covariance** of supplier TFP growth in the current and prior periods (persistence)
- ii) **Lagged dependent variable**, account for mean reversion in industry TFP
- iii) **China imports penetration**, could affect measured TFP
- iv) **Dropping Computers**; not drive solely by this sector, but evidence it plays a major role
- v) **Fixed Input-Output table** instead of time-varying one

# Conclusion

- Innovation in an industry relies on *complementary* innovations in linked industries (input suppliers)
- Unbalanced technology advances across different industries holds up aggregate productivity
  - Creates endogenous innovation bottlenecks across input-output linkages

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## Discussion

- Not necessary causal
- Does not address questions of efficiency and misallocation
  - Possibly underlying frictions
- Extend to larger historical time period

# Appendix